

**TS Question Paper 2017**  
**Class 12 Physics, Paper-II**

**Time allowed : 3hours**

**Maximum Marks : 60**

**SECTION-A**

**(10 × 2 = 20)**

**Note:i) Answer all the questions.**

**ii) Each question carries two marks.**

**iii) All are very short answer type questions.**

**1. What is the importance of oersted's experiment?**

Ans. Importance of oersted's experiment:

- It helps in showing the relationship between the electricity and magnetism.
- Importance of this experiment is that by the help of this experiment we can use many applications like motors which is based on tis phenomenon.

**2. A concave mirror produces an image of a long vertical pin, placed 40 cm from the concave mirror , at the position of the object . Find the focal length of the mirror.**

Ans : Using the mirror formula,

$$1/f = 1/u + 1/v$$

given,  $u = v = 40\text{cm}$

$$\text{so, } 1/f = (1+1)/40$$

$$1/f = 1/20$$

$$f = 20$$

therefore focal length is 20 cm

### **3. What happens to compass needles at the earth's poles?**

Ans. Horizontal component of earth's magnetic field is zero at the earth's poles. So the compass needle is move in many direction.

### **4. Magnetic lines form continuous closed loops. Why ?**

Ans. Magnetic lines forms the continuous loops because magnetic fields lines present the continuous curves starts from the north pole and ends in the south pole and inside the magnet they moves from the south to the north pole.

### **5. What is meant by wattless component of the current ?**

Ans. It is also known as the idle current. It is defined as the wattless current when the average power used in such circuit corresponds to zero.

### **6. Give two uses of infrared rays.**

Ans : We use infrared radiation in many ways

- thermal imaging cameras
- remote controls

## 7. What is "work function"?

Ans : It is defined as the minimum quantity of energy which is required to remove an electron to infinity from the surface of a given solid, usually a metal.

## 8. State heisenbergs ' s uncertainty Principle.

Ans : Heisenberg's uncertainty principle :- Heisenberg sir states that it is impossible to measure simultaneously the exact position and momentum of subatomic particle. That is, the more exactly the position is determined, the less known the momentum, and vice versa.

According to him, " the product of uncertainty in position and uncertainty in momentum be always greater than equal to  $h/4\pi$  "

e.g.,  $\Delta x \cdot \Delta p = h/4\pi$

Here  $\Delta x$  is uncertainty in position

$\Delta p$  is uncertainty in momentum .

## 9. What is p- type semiconductor ? What are the majority and minority charge carriers in it ?

Ans. When the intrinsic or pure semiconductor is mixed with the some impurity, then it is known as p-type semiconductor.

Majority charge carriers are positive charges i.e., holes

Minority charge carriers are negative charges i.e., electrons

**10 . Mention the basic methods of modulation.**

Ans: Modulation is nothing but, a carrier signal that varies in accordance with the message signal. Modulation technique is used to change the signal characteristics. Basically, the modulation is of following two types:

- Analog Modulation
- Digital Modulation

## SECTION- B

 $(6 \times 4 = 24)$ 

**Note: i) Answer Any Six of the following questions.**

**ii) Each question carries four marks.**

**iii) All are short answer type questions**

**11. Why does the setting sun appear red ?**

Ans : So, sunlight nears the horizon which passes through thick layers of air and through the larger distance of atmosphere. Hence, most of blue light is scattered away and longer wavelengths reach our eyes which gives reddish color of sun.

**12. Discuss the intensity of transmitted light when a polaroid sheet is rotated between two crossed polaroids.**

Ans - Let  $P_1$  and  $P_3$  are cross polaroids.  $P_2$  is the another polariser that is placed in between  $P_1$  and  $P_3$  and is rotated.

Let  $I_0$  be the intensity of polaroid light after passing through first polariser  $P_1$ .

Then the intensity of light passing through second polariser  $P_2$  will be

$$I = I_0 \cos^2 \theta$$

where  $\theta$  is the angle between pass axes of  $P_1$  and  $P_2$ .

Since  $P_1$  and  $P_3$  are crossed and the angle between the pass axes

of  $P_2$  and  $P_3$  will be  $[(\pi/2) - \theta]$ .

Hence intensity of light emerging from  $P_3$  will be

$$I = I_0 \cos^2 \theta \cos^2[(\pi/2) - \theta] = I_0 \cos^2 \theta \sin^2 \theta = (I_0/4) \sin^2 2\theta$$

therefore the transmitted intensity will be maximum when  $\theta = \pi/4$

### **13. State and explain Coulomb's inverse square law in electricity.**

Ans. Coulomb's inverse law in electrostatics states that the force of interaction between the two points is directly proportional to the multiplication of magnitudes of charges and inversely proportional to the square of the distance between them. Mathematically it is represented as the.,

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$$

### **14. Explain parallel combination of capacitor. derive the formula for equivalent capacitance in parallel combination.**

Ans. Parallel combination-

It is defined as the three capacitors, with one plate of each capacitor connected to one side of the circuit and the other plate is connected to the other side.

Formula for the equivalent capacitance in parallel combination:

total charge stored  $Q$  is the sum of all the charge in different capacitors,

$$Q = Q_1 + Q_2 + Q_3 \dots \dots \dots \text{equation 1}$$

Use the relation,  $Q = C_p V$

$$\text{So, } Q_1 = C_1 V$$

$$Q_2 = C_2 V$$

$$Q_3 = C_3 V$$

After putting these values in the equation 1 we get,

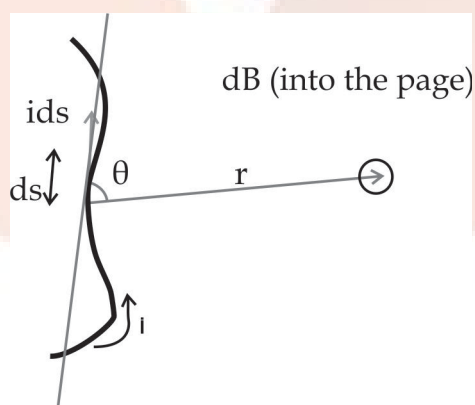
$$C_p V = C_1 V + C_2 V + C_3 V$$

After the simplified the above equation we get the equivalent for the parallel combination,

$$C_p = C_1 + C_2 + C_3$$

### 15. State and explain Biot Savart law.

Ans:-Biot-Savart's law is an equation that gives the magnetic field produced due to a current carrying segment. This segment is taken as a vector quantity known as the current element.



Biot Savart Law states that

The magnetic intensity  $dH$  at a point A due to current  $I$  flowing through a small element  $dl$  is

1. Directly proportional to current ( $I$ )
2. Directly proportional to the length of the element ( $dl$ )
3. Directly proportional to the sine of angle  $\theta$  between the direction of current and the line joining the element  $dl$  from point A.
4. Inversely proportional to the square of the distance ( $x$ ) of point

$$dH = \frac{\mu_0 \mu_r}{4\pi} x I dl \sin \theta / x^2$$

$$dH = k x I dl \sin \theta / x^2$$

$$dH \propto x I dl \sin \theta / x^2$$

where  $k$  is constant and depends on the magnetic properties of the medium.

$$K = \mu_0 \mu_r / 4\pi$$

$\mu_0$  = absolute permeability of air or vacuum and its value is  $4 \times 10^{-7} \text{ Wb/A-m}$

$\mu_r$  = relative permeability of the medium.

**16. A pair of adjacent coils has a mutual inductance of 1.5 H. if the current in one coil changes from 0 to 20 A in 0.5 sec.**



**what is the change of flux linkage with the other coil ?**

Ans : In the above question, a pair of adjacent coils

Mutual Inductance,  $M = 1.5H$

Current in the coil,  $I = 20A$

Time =  $0.5\text{sec}$

Change in flux =  $1.5 \times 20$

=  $30H$

**17. Write a short note on DeBroglie's explanation of bohr's second postulate of quantization.**

Ans. DeBroglie's explanation of Bohr's second postulate of quantization:

Those waves which forms the nodes at a terminal point with integral multiple of wavelength is propagate and survive. It is considered as a particle wave, when an electron is orbiting in circular orbit. From the DeBroglie's equation we obtained

$$\lambda = h/(mv)$$

The help of this equation we get

$$mvnrn = nh/(2\pi)$$

This equation proved that wave-particle duality is cause of quantised energy state.

### 18. Define NAND and NOR gates . Give their truth tables.

ans:-The NAND gate is a combination of an AND gate and NOT gate. They are connected in cascade form. It is also called Negated And gate. The NAND gate provides the false or low output only when their outputs is high or true. ... The function completeness means any types of gates can be implemented by using the NAND gate.

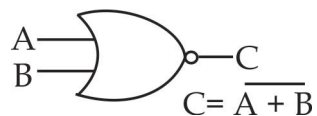
#### NAND GATE



#### Truth Table

| INPUT |   | OUTPUT   |  |
|-------|---|----------|--|
| A     | B | A NAND B |  |
| 0     | 0 | 1        |  |
| 0     | 1 | 1        |  |
| 1     | 0 | 1        |  |
| 1     | 1 | 0        |  |

OR gate followed by a NOT gate in a cascade is called a NOR gate. In other words, the gate which provides a high output signal only when there are low signals on the inputs such type of gate is known as nor gate



## NOR GATE

### Truth Table

| INPUT |   | OUTPUT  |  |
|-------|---|---------|--|
| A     | B | A NOR B |  |
| 0     | 0 | 1       |  |
| 0     | 1 | 1       |  |
| 1     | 0 | 1       |  |
| 1     | 1 | 0       |  |

## SECTION –C

(2 × 18 =16)

**Note: i) Answer ANY TWO of the following questions.**

**ii) Each question carries Eight marks.**

**iii) All are long answer type questions**

**19. What is doppler effect ? obtain an expression for the apparent frequency of sound heard , when the source is in motion with respect to an observer at rest .**

Ans. Doppler effect - it referred to the change in a frequency of a wave on behalf of the observer which is moving according to the the wave source.

An expression for the apparent frequency of sound heard:

Let the velocity of source=  $V_s$

Distance travelled in time period,  $T = V_s T$

Apparent wave ,  $\lambda' = \lambda - V_s T$

If frequency is  $v$ ,.

$$\lambda = \lambda' - V_s T$$

$$\lambda = \lambda' - V_s / T$$

$$\lambda = V - V_s / V$$

$$V = v \lambda$$

Apparent frequency heard by listener is  $v'$

$$v' = v / \lambda'$$

$$= vV / V - V_s$$

Similarly if source moves away from listener,

$$v' = V / \lambda'$$

$$= vV / V + V_s$$

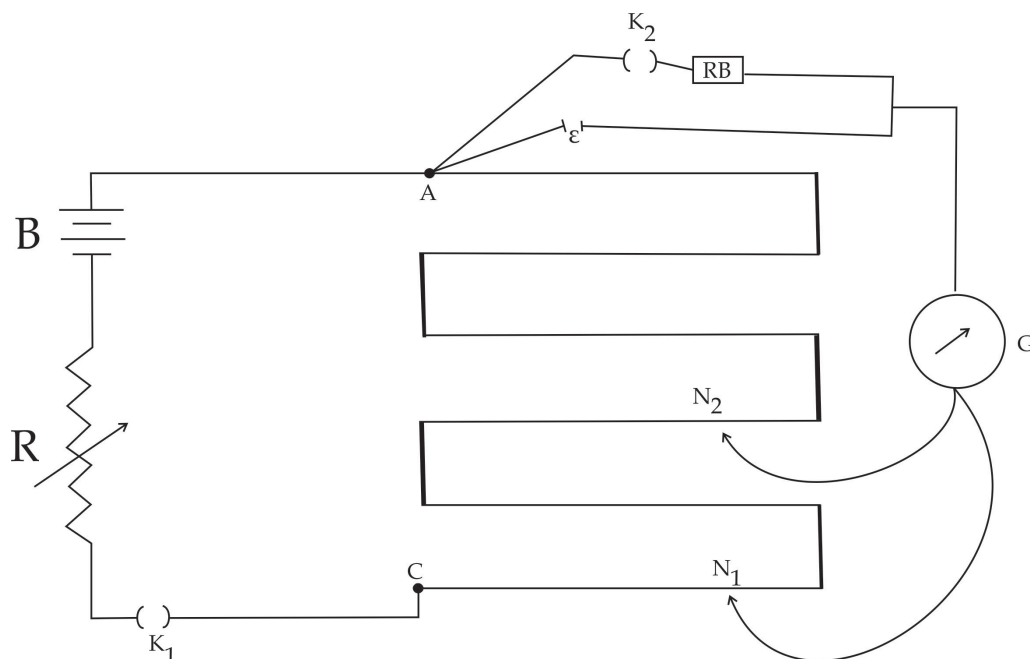
this above equation is the apparent frequency of sound heard.

**20. State the working principle of potentiometer . Explain with the help of a circuit diagram , how the potentiometer is used to determine the internal resistance of the given primary cell.**

**A battery of emf 2.5 v and internal resistance "r" is connected in series with a resistor of 45 ohm through an ammeter of resistance 1 ohm . The ammeter reads a current of 50 mA . calculate the internal resistance r .**

Ans :

Principle of working: The voltage drop along the wire is directly proportional to the length of the wire. The potentiometer works without drawing any current from the voltage source.



We can also use a potentiometer to measure internal resistance of a cell. For this the cell (emf  $E$ ) whose internal resistance ( $r$ ) is to be determined is connected across a resistance box through a Key  $K_2$ , as shown in the figure.

With key  $K_2$  open, balance is obtained at length  $L_1$  ( $AN_1$ ).

Then,

$$E = \Phi L_1$$

When key  $K_2$  is closed, the cell sends a current ( $I$ ) through the resistance box ( $R$ ). If  $V$  is the

terminal potential difference of the cell and balance is obtained at length  $L_2$  ( $AN_2$ ), then

$$V = \Phi L_2$$

So, we have  $E/V = L_1/L_2$

But,  $E = I(r + R)$  and  $V = IR$ . This gives

$$E/V = (r + R)/R$$

On comparing the above equations, we get,

$$(R + r)/R = L_1/L_2$$

$$r = R \left( \frac{L_1}{L_2} - 1 \right)$$

Total resistance of the circuit =  $45 + 1 + r = (46 + r)$  ohm

Current =  $50 \times 10^{-3}$  A

Emf = 2.5V

$\Rightarrow$  Resistance = emf/current

$\Rightarrow (46 + r)$  ohm =  $2.5 \text{ V} / 50 \times 10^{-3} \text{ A} = 50$  ohm

$\Rightarrow r = (50 - 46)$  ohm

= 4 ohm.

**21. Explain the principle and working of a nuclear reactor with the help of a labelled diagram .**

Ans :-A nuclear reactor works on the principle of achieving control chain reaction in Uranium (238) enriched with Uranium (235) generating large amounts of energy.

The reactor consists of fuel, moderators, control rods, radiation shielding and coolants.

**Fuel:** In reactor, fuel is fabricated in form of thin and long cylindrical rods together known as fuel assembly. The rods are surrounded by coolant for transfer of heat produced in them.

Natural uranium is used as nuclear fuels.

Moderators are used to slow down the velocity of neutrons. E.g. graphite, heavy water .

Control rods are used to control the fission rate by absorbing neutrons. Cadmium, Boron are used to serve the purpose.

Radiation shielding by the utilisation of steel, lead, concrete etc. are provided to absorb and reduce the intensity of beta and gamma radiations emitted during fission, to such low levels that it does not become harmful to operating personnel.

